

GOODS MARKET

The aim of this chapter is to integrate the money and commodities market, with primary emphasis on the mechanics of their interaction. We also show how alternative possible interactions between money and commodities may affect crucial macroeconomics variables such as income, employment and the rate of interest. We consider the possible effects of fiscal and monetary policy, the fundamental tools of economics stabilization.

The Goods market also common known as the commodity or the product market is one where output (Y) is produced and later consumed by the economic agents. It is from the goods market that we have the national income identity, which we defined using the equation $Y = C + I + G$

To develop this relationship we use a basic model called the Keynesian cross. In *The General Theory* Keynes proposed that an economy's total income was in the short run, determined largely by the desire to spend by households, firms and the government. The more people want to spend, the more goods and services firms can sell. The more firms can sell, the more output they will choose to produce and the more workers they will choose to hire.

We develop a commodities market equilibrium function called the IS or investment-saving function. We make an assumption the interest rate is allowed to vary and to effect investment i.e. investment is inversely related to the interest rate. $I = I(i)$. This condition means that investment will increase as interest rate fall and vice versa.

The IS function

Recall that in the previous chapter, equilibrium in the goods market comprising consumer, investor and government sector required that

$$S + T = I + G$$

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Leakages injections

The S = I equilibrium condition of the goods market

Assuming an economy that does not involve in foreign trade, using the income approach

National Income (Y) is expressed as $Y = C + S + T$.

Using the expenditure approach, it expressed as $Y = C + G + I = AD$

Where: Y = National Income C = Consumption S = Savings T = Tax

G = Government Expenditure and I = Investment

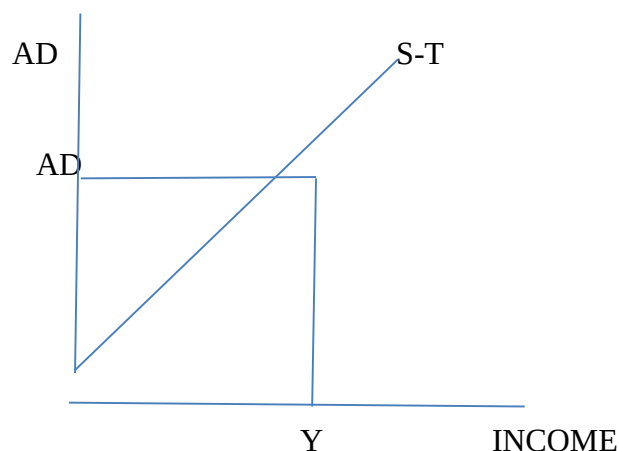
Equilibrium in the goods market requires that aggregate income (output) be equal to aggregate expenditure. Therefore, the goods market equilibrium can be expressed as

$$C + S + T = C + G + I$$

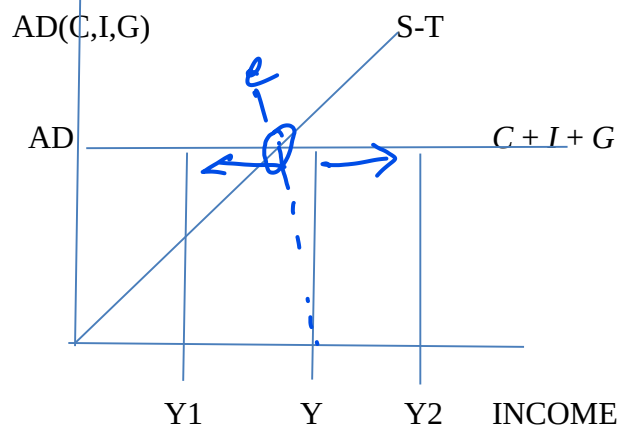
$$S + T = G + I$$

Assuming that $G = T$, then equilibrium is achieved when $I = S$.

The $I = S$ is the equilibrium condition in the goods market, meaning that the economy should be at equilibrium at any point along $S = I$ line.



At equilibrium, all variables are fixed. This means that $AD = C + I + G$ NB/ PUT MEAN



AD -

The above figure shows condition for equilibrium income.

At Y_2 , there is excess supply hence disequilibrium exists in the economy.

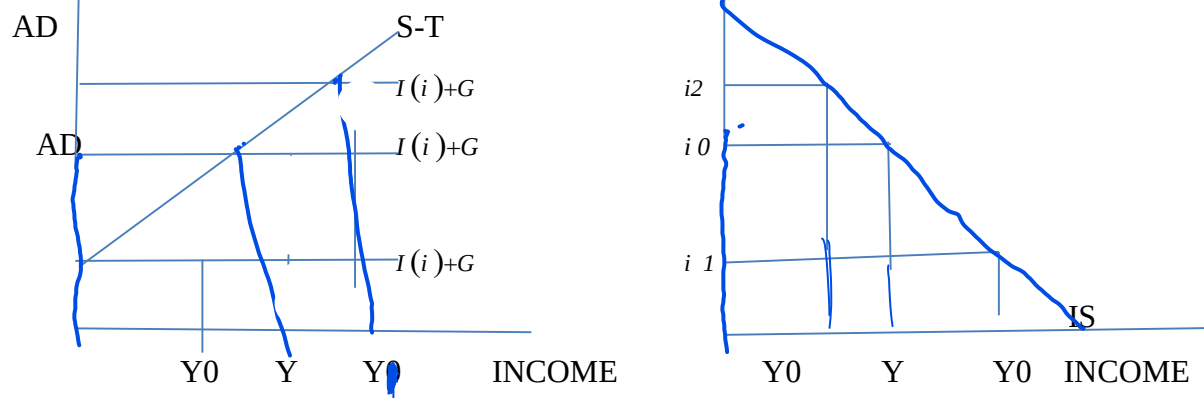
At Y_1 , demand is greater than supply and disequilibrium exists in the economy.

These situations of disequilibrium are normally short-lived as there is tendency for income to move towards Y^* .

When savings is equal to investment, the addition to demand due to increase in investment is equal to subtraction from demand due to less spending by households

We know that aggregate demand is given as $AD = C + I + G$

If both government expenditure and investment are given (Constant) then changes in AD will be due to changes in consumption. The following diagram shows equilibrium incomes in a closed economy at different levels of Aggregate expenditure.



In panel a, with interest rate i_0 equilibrium income is at level Y_0 since leakages equal

Injection at that income level $[S + T = I(i_0) + G]_0$.

- However a lower interest rate i_1 will increase investment spending and shift

$I(i)_0 + G$ to $I(i)_1 + G$. Income will rise from Y_0 to Y_1 .

- Similarly a higher interest rate i_2 will cause $I(i)_0 + G$ to shift downwards to

$I(i)_2 + G$, resulting in a lower income level Y_2 .

- The inverse relationship between the interest rate and income is plotted on the IS curve in panel (b), which shows the various combination of i and Y that produce equilibrium in the real sector.

- If the economy is situated to the right of the IS curve (such as at point R in panel b), then disequilibrium exists because $S + T > I + G$. The income level is “too high” for

the associated interest rate and the high income level “too much” saving and tax. Alternatively, for the income level associated with R, the interest rate is too high and is thus chocking off investment. Income will fall until the IS curve is reached through cut backs of production because of unintended inventory accumulation at the higher level of income.

- To the left of the IS curve, such as at point U, $S + T < I + G$ and there is upwards pressure on the income level.

We have seen that equilibrium in the goods market require that $S = I$. all factors that cause changes will ^{be} Consumption, Savings and Investments will therefore influence the equilibrium position

Suppose the consumption (C) and investment functions are given as

$$C = \alpha + \beta Y; 0 < \beta < 1$$

$$I = I_0 - kr; 0 < k < 1$$

Where; I_0 = autonomous investment; α = autonomous consumption; r = interest rate and β & k = parameters

1. Using the expenditure approach of deriving the IS equation, and we assume absence of government

$$Y = C + I$$

$$Y = \alpha + \beta Y + I_0 - kr$$

$$Y - \beta Y = \alpha + I_0 - kr$$

$$Y(1 - \beta) = \alpha + I_0 - kr$$

$$kr = \alpha + I_0 - Y(1 - \beta)$$

Hence

$$r = \alpha/k + I_0/k - Y(1 - \beta)/k$$

2. Using the savings approach of deriving the IS equation we recall that $S = sY$; where s is a parameter and $0 < s < 1$

At equilibrium $S = I$; therefore

$$sY = I_0 - kr$$

$$kr = I_0 - sY$$

Hence

$$r = I_0/k - S_y/k = \text{IS equation}$$

Example

(a) The commodity and money markets of a certain economy are given as follows:

Commodity market:

$$Y = C = I$$

National Income

$$C = 100 + 0.6Y$$

Consumption function

$$I = 1800 - 2.5r$$

Investment function

Derive the IS equation and explain what it implies

Solution:

Substitute the Consumption and Investment Functions

We are given:

$$C = 100 + 0.6Y$$

$$I = 1800 - 2.5r$$

Substituting these into the equilibrium equation:

$$Y = (100 + 0.6Y) + (1800 - 2.5r)$$

$$Y = 100 + 0.6Y + 1800 - 2r$$

$$Y - 0.6Y = 1900 - 2r$$

$$0.4Y = 1900 - 2r$$

$$Y = 4750 - 5r$$

Interpretation of the IS Equation

1. **Slope (- 5):** The negative slope of the IS curve indicates an **inverse relationship** between the interest rate (r) and national income (YYY). As r increases, investment decreases, leading to a decline in national income.

National Income

$$\text{slope} = \frac{\Delta Y}{\Delta r}$$

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2. **Intercept (4750):** This represents the level of income when the interest rate (r) is zero.
 3. **Implication:** The IS curve shows all combinations of **income (Y) and interest rate (r)** that maintain equilibrium in the goods market. It reflects how investment and consumption respond to changes in interest rates.

MULTIPLIER CONCEPT

The Multiplier Concept is an economic principle that explains how an initial change in spending (investment expenditure or government expenditure) leads to a larger overall change in national income. It is based on the principle that one person's spending becomes another person's income, leading to a chain reaction of increased economic activity.

zero sum game

Types of Multipliers

1. Investment Multiplier – Shows how an increase in investment leads to a larger increase in income and output.

income > Govt

$$k = \frac{1}{1-MPC}$$

where k is the multiplier and MPC (Marginal Propensity to Consume) is the fraction of additional income that people spend on consumption.

2. Fiscal Multiplier – Measures the effect of government spending or tax changes on GDP.
3. Money Multiplier – Explains how an increase in the money supply (through bank deposits and lending) results in a greater expansion of the economy.
4. Employment Multiplier – Describes how an increase in jobs in one sector leads to job creation in related industries.

Key Factors Influencing the Multiplier Effect

- Marginal Propensity to Consume (MPC): Higher MPC leads to a greater multiplier effect.
- Marginal Propensity to Save (MPS): Higher MPS leads to a lower multiplier effect.
- Leakages: Savings, taxes, and imports reduce the impact of the multiplier.
- Consumer Confidence: If people are uncertain about the future, they may save instead of spending, reducing the multiplier effect.

Practical Applications

- Government Policy: Used to assess the impact of fiscal stimulus or austerity measures.

- Business Decisions: Helps in predicting how investment in new projects will influence economic growth.
- Monetary Policy: Central banks use it to measure the effect of money supply changes on the economy.

INFLATIONARY GAP

An inflationary gap occurs when aggregate demand (AD) exceeds the economy's potential output (full employment level of GDP). This leads to increased price levels and demand-pull inflation.

Causes of Inflationary Gap

- Excessive government spending
- Increase in consumer or business confidence leading to higher consumption and investment
- Expansionary monetary policies (e.g., low-interest rates, increased money supply)
- High exports due to favorable international trade conditions

Effects of an Inflationary Gap

- Rising inflation due to demand exceeding supply
- Overheating of the economy, leading to potential asset bubbles
- Labor shortages, increasing wages and production costs
- Higher interest rates as central banks attempt to control inflation

Policy Measures to Reduce an Inflationary Gap

- **Contractionary Fiscal Policy:** Government reduces spending or increases taxes
- **Contractionary Monetary Policy:** Central bank raises interest rates or reduces the money supply

- **Supply-Side Policies:** Increase production capacity through investments in technology, infrastructure, and labor force training.

DEFLATIONARY GAP

A deflationary gap occurs when aggregate demand is lower than the economy's full employment level of output, leading to unemployment and lower price levels.

Causes of Deflationary Gap

- Reduced consumer and business spending due to low confidence
- High taxation or reduced government spending (austerity measures)
- Contractionary monetary policies (e.g., high-interest rates)
- Declining exports due to weak global demand

Effects of a Deflationary Gap

- Falling prices (deflation) and reduced business revenues
- Increased unemployment due to lower production
- Reduced economic growth and investment
- Potential for a recession if the gap persists

Policy Measures to Reduce a Deflationary Gap

- **Expansionary Fiscal Policy:** Government increases spending or reduces taxes
- **Expansionary Monetary Policy:** Central bank lowers interest rates or increases money supply
- **Job Creation Programs:** Government investment in infrastructure and public projects to boost employment

Equilibrium In The Money Market: The Lm Curve.

The LM curve plots the relationship between interest rate and the level of income that arises in the market for money balances. To understand this relationship, we begin by looking at a theory of the interest rate; called the theory of liquidity preference. In his book “General Theory”, Keynes offered his view of how the interest rate is determined in the short run. He argued that interest rate adjusts to balance the supply and demand for the economy’s most liquid asset-money.

Supply for real money balances

If M stands for the supply of money and P stands for the price level, Then M/P is the supply of real money balances. The theory of liquidity preferences assumes there is a fixed supply of real money balances. The money supply M is an exogenous policy variable chosen by a central bank. The price level P is also an exogenous variable in this model. These assumptions imply that the supply of real balances is fixed, in particular does not depend on

Keynesian Theory of Demand for Money (Liquidity Preference Theory)

According to Keynes there are three main motives for holding money.

- i) Transactions motive
- ii) Speculative Motive
- iii) Precautionary Motive

(i) Transactions Demand for Money

This is the amount of money that households, firms and the government hold, for future exchange of goods and services. These agencies hold money because income (money receipts) and expenditure flows are not synchronized in time. Expenditure tends to be continuous while receipts are discrete. Money is therefore required to bridge the gap between receipts and disbursements. The amount of money held for transaction purposes depend on:

The level of income-the higher the income the higher the amount held

- Individuals spending habits
- The frequency of payment of an individual

- Availability of credit facilities
- Time interval in-between income receipts
- The rate of inflation

Money is held at an opportunity cost of the interest it would earn if it were invested.

Holding spending habits and interval between income receipts constant, the higher the income, the higher the amount of money held for transaction motive.

Transactions demand for money is given by

$$M_t = L_t(Y)$$

Where; Y = Level of income, M_t = Total amount of money demanded for transaction motive and L_t = Liquidity function corresponding to the money income

OR

$$M_t = kY$$

Where $k = L_t$ and $0 < k < 1$

In the post Keynesian theory therefore, transaction demand for money is a function of income (Y) and interest rate (r) i.e. $M_t = f(Y, r)$

Precautionary Demand for Money

This is the amount of money held to provide for unexpected expenditures. According to Keynes precautionary demand is determined by an individual's income and institutional factors, which he considered fixed in the short run, so that;

$$M_t = kY,$$

Where M_t is combined transaction and precautionary demand for real cash balances.

Speculation Demand for Money

It refers to the amount of money held to take advantage of profitable opportunities that may arise in financial markets. Whereas the transactions and precautionary motives of demand for money look at money as a medium of exchange, the speculative motive looks at it as store of value.

People hold money for speculation purposes due to uncertainty about the future rate of interest. A fall in the interest rate leads to an increase in the speculation demand for money.

The concept of velocity of money

- Velocity of circulation of money is the rate at which money stock is used to make transactions for final goods and services.

- This was put forward by Irving Fisher, in his version of the quantity theory of money

- Fisher summarized the quantity theory of money in the following equation:

$$MV \equiv PT$$

Where: M = nominal stock of money in the economy

V = the transactions velocity of circulation of money

P = the general price level of all transactions

T = the number of transactions that take place during the time period

- Both MV and PT measure the total value of transactions during the time period hence are always identical.

- For example if we suppose that in a given economy the number of transactions (T) = 100 and the average price (P) is 10/€, then $PT = 1000/€$. If the stock of money is 100/€, then the number of times it changes hands (V) = 10.

- Therefore given any three of the components of the identity $MV = PT$, it is possible to compute the value of the remaining component.

Classical theory of money supply

- Classical economists assumed that both V and Q are constant.

- V is constant because of people's stable behavior of holding money.

- Q is constant because the economy is assumed to be at full employment.

- If those assumptions hold, then from the exchange equation ($MV = PQ$), the price level must be proportional to money supply (M).

- In the short run the real GNP will not change hence